

HTML and understanding HTML document structure and basic tags

1. What is HTML?

HTML stands for **HyperText Markup Language**. It is the standard language used to create and structure webpages. Web browsers (like Chrome, Firefox, Safari) read HTML code and display it as websites.

2. Why learn HTML?

- ✓ It's the foundation of all websites.
- ✓ You need html to build pages before adding **Styles (using CSS language) and functionality (JavaScript)**
- ✓ Knowing html helps you understand how websites work behind the scenes.

Basic terminology

Elements: - A piece of content in a webpage (like paragraph, heading, or image).

Tag: - Special keyboards inside angle brackets like <P> or <h1> that define elements.

Attribute: - Extra information added to tags like **class, id, href, src** etc. these all are attributes in tags.

HTML is the foundational language of the web, the structure upon which everything else is built. When your browser loads a webpage, it's primarily reading an HTML document to understand what content to display and how it's organized. This isn't just a collection of text; it's a carefully structured document that tells the browser how to render text, images, videos, and interactive elements.

Every HTML document starts with a Document Type Declaration, or DOCTYPE. This isn't an HTML tag itself, but rather an instruction to the web browser about which version of HTML the page is written in. For modern web development, we use HTML5, and its DOCTYPE is remarkably simple:

```
<!DOCTYPE html>
```

This declaration must be the very first thing in your HTML file, before any other tags. It ensures that browsers render your page in "standards mode," which means they follow modern web standards rather than attempting to emulate older, often inconsistent, rendering behaviors. Without it, browsers might switch to "quirks mode," leading to unexpected layout and styling issues. Always include it.

3. The Root Element:

Immediately after the declaration comes the tag. This is the root element of every HTML page. All other content for your webpage, everything you see and much of what the browser uses internally, lives inside this tag. Think of it as the container for your entire document.

It's common practice to include a lang attribute on the tag. This attribute declares the primary language of the document, which is important for accessibility tools (like screen readers) and search engines. For English content, you'd use lang="en".

```
<!DOCTYPE html>
<html lang="en">
  <!-- ALL other HTML content goes here -->
</html>
```

4. The Document Head:

Inside the element, the first major section you'll find is the. This section contains metadata about the HTML document itself. Metadata is data about data—information that isn't directly displayed on the page as content but provides crucial instructions and details to the browser, search engines, and other web services.

`<meta>`

tags: These provide various types of metadata. A critical one is `meta charset="UTF-8"`, which defines the character encoding for the document. UTF-8 supports almost all characters and symbols in the world, ensuring text displays correctly across different languages and platforms. Another common one is `meta name="viewport" content="width=device-width, initial-scale=1.0"`, which is vital for responsive design, instructing browsers how to control the page's dimensions and scaling on different devices.

`<title>` tag: This defines the **title of the document**, which appears in the browser tab, window title bar, or as the default name when bookmarking a page. It's a key factor for user experience and search engine optimization (SEO).

`<link>` tags: These are used to link external resources to your HTML document. The most common use is linking to external CSS stylesheets, for example: `<link rel="stylesheet" href="styles.css">`.

`<script>` tags: While JavaScript typically goes just before the closing `</body>` tag for performance reasons, sometimes you'll see `<script>` tags in the `<head>`, particularly if the script needs to run before the page content loads or if it's linking an external script.

Here's an example of a typical `<head>` section:

```
>>>> Basic html structure or basic html code <<<<<
```

```
<!DOCTYPE html>
<html lang="en">
<head>

  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Awesome Website</title>
  <link rel="stylesheet" href="styles.css">
  <script src="my-script.js"></script>
</head>
<body>
  <!-- Page content will go here -->
</body>
</html>
```

5. The Visible Part: <body>

What <body> Does

The `<body>` element contains **everything the user sees** on the page:

- Text, headings, paragraphs.
- Images, links, buttons, forms.
- Anything interactive or visual.

Example:

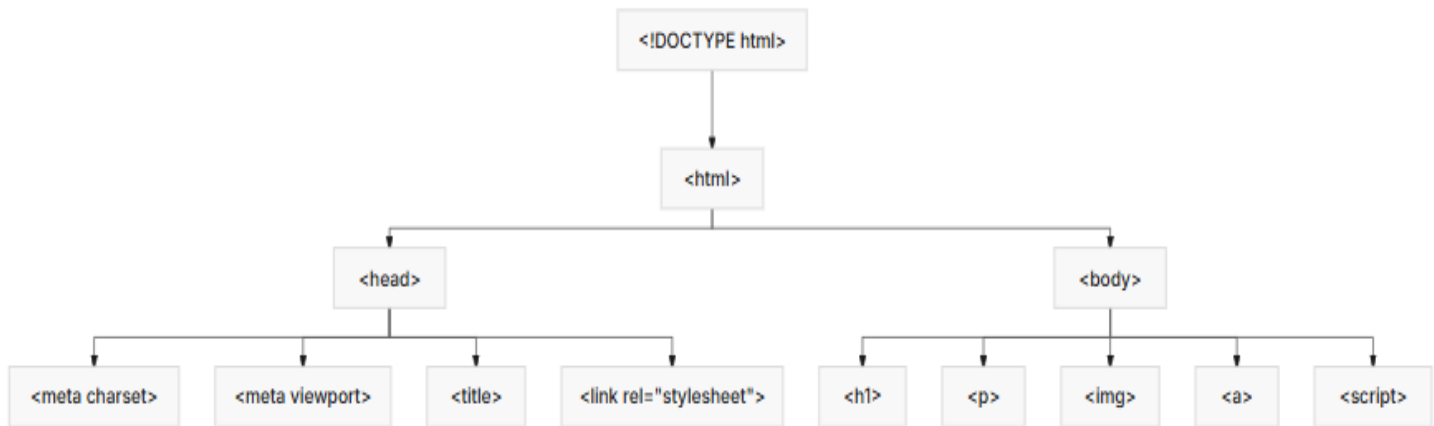
```
<body>
  <h1>Welcome</h1>
  <p>This is visible on the screen.</p>
</body>
```

- Without `<body>`, the user would see **nothing**, even if the HTML structure is correct.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Awesome Website</title>
</head>
<body>
  <h1>Welcome to My Website!</h1>
  <p>This is a paragraph of content.</p>
</body>
</html>

```



Short revise point of html structure

<!DOCTYPE html> - tell the browser this is an html5 document.

<html> - is the root of the HTML page.

<head> - contains information about the page (not show on screen or browser screen).

<title> - sets the title seen on the browser tab.

<body> - contains everything visible on the webpage.

6. HTML Basic Tags: -

Heading tags in html (h1 to h6)

Headings help you organize content into sections. Html provides 6 levels of heading: -

<h1> - Main Heading (biggest)

<h2> - Subheading

<h3> - smaller subheading

<h4> <h5> <h6> - even smaller headings

```
<h1>The Grand Title of My Webpage</h1>
  <h2>A Major Section of Content</h2>
    <h3>Another Sub-Section</h3>
      <h4>A Smaller Point Within the Sub-Section</h4>
        <h5>A smaller heading than h4 </h5>
          <h6>a smallest heading in html </h6>
```



Tip:

Use only one `<h1>` per page (usually for the page title).

Use headings to structure your content, not to make text look big (that's CSS's job).

7. Paragraphs in html

Paragraphs are written using the `<p>` tag

```
<p>This is a paragraph of text.</p>
```

Browsers add a bit of space above and below each paragraph automatically.

Why It Matters

- It tells the browser and assistive tools, "this is a block of text."
- Proper use improves readability and accessibility.

Line Breaks

If you want to break a line without starting a new paragraph, use the `<br>` tag.

```
<p> this is paragraphs1 <br> and this is the second line </p>
```

➤ Task 1: use heading and paragraph and br tag and create basic website

